

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
PGA BW	PGA bandwidth for different inverting gain	Gain = -3		-	GBW/4	-	MHz
		Gain = -7		-	GBW/8	-	
		Gain = -15		-	GBW/16	-	
en	Voltage noise density	at 1 KHz	output loaded with 4 kΩ	-	360	-	nV/√ Hz
		at 10 KHz		-	160	-	
I _{DDA} (OPAMP)	OPAMP consumption from V _{DDA}	Normal mode	no Load, quiescent mode, follower	-	1.6	2.8	mA
		High- speed mode		-	1.8	3	
		Normal mode	OPAMPint, follower	-	0.8	1.6	
		High- speed mode		-	1	1.8	

1. R_{LOAD} is the resistive load connected to V_{SSA} or to V_{DDA} .
2. R_2 is the internal resistance between the OPAMP output and the OPAMP inverting input. R_1 is the internal resistance between the OPAMP inverting input and ground. The programmable gain amplifier (PGA) is calculated as $1 + R_2/R_1$.

5.3.22 Timer characteristics

The parameters given in Section 5.3.22, Table 61, and Section 5.3.22 are specified by design, not tested in production.

Refer to Table 46. I/O static characteristics for details on the input/output alternate function characteristics (output compare, input capture, external clock, PWM output).

Table 60. TIMx characteristics

Symbol	Parameter	Conditions	Min	Max	Unit ⁽¹⁾
t _{res} (TIM)	Timer resolution time	-	1	-	t _{TIMxCLK}
		f _{TIMxCLK} = 144 MHz	6.9	-	ns
f _{EXT}	Timer external clock frequency on CH1 to CH4	-	0	f _{TIMxCLK} /2	MHz
		f _{TIMxCLK} = 144 MHz	0	72	
ResTIM	Timer resolution	TIMx (except TIM2/TIM5)	-	16	bit
		TIM2/TIM5	-	32	
t _{COUNTER}	16-bit counter clock period	-	1	65536	t _{TIMxCLK}
		f _{TIMxCLK} = 144 MHz	0.007	455.1	μs
t _{MAX_COUNT}	Maximum possible count with 32-bit counter	-	-	4, 294, 967, 296	t _{TIMxCLK}
		f _{TIMxCLK} = 144 MHz	-	29.826	s

1. TIMx, is used as a general term in which x stands for 1, 2, 5, 6, 7, 8, 12, 15, 16, 17.